BellaBilling™

Applying Google's AI to a Human-Approved Billing Workflow

Introduction

Small service businesses spend an incredible amount of time on repetitive tasks and administrative work that happens after the job is already finished. One of the most common examples is invoicing. The work is complete and now someone has to remember the details, open the billing software, re-create the job from memory, generate the invoice and send it manually.

As the founder of ZaderFlo Systems LLC, I've spent the last several months designing AI-powered business systems specifically for service professionals. BellaBilling™ is one product within a larger ecosystem of specialized business agents built to reduce administrative work while keeping business owners in complete control. The Bella by ZaderFlo Ecosystem.

Rather than creating a prototype specifically for this competition, I used this capstone as an opportunity to apply the concepts from Google's AI Intensive to a real product already under active development. The result is a production-oriented AI billing coordinator that transforms a natural conversation into a customer ready invoice through a human-approved workflow.

BellaBilling™ is not designed to replace the business owner. It replaces the paperwork.

The problem

For many service businesses, invoicing is a surprisingly difficult process.

After completing a job, the owner often has to stop what they're doing, remember the work that was performed, manually type customer information, calculate totals, generate a PDF, and finally email the invoice. And let's not even *start* on them keeping track of paid and unpaid invoices.

The delay isn't caused by technical difficulty.
It's caused by repetitive tasks and administrative friction.

All of those small interruptions and frictions throughout the day reduce the amount of time business owners can spend serving customers.

I wanted to explore whether an AI agent could remove that administrative burden without removing the business owners' own authority over the final invoice.

Why an agent

One of the biggest lessons I took away from Google's AI Intensive was that *not every problem should be solved with a chatbot.*

BellaBilling™ is not a conversational assistant that simply answers questions.

It is an agent operating inside a structured workflow.

The conversation provides context. The LLM extracts structured billing information.

Automation coordinates the workflow. The business owner reviews the result.

Only after explicit **human approval** does the system generate and deliver the final invoice.

Throughout the course, the concept of **Agent = Model + Harness** became one of the most important design principles for me.

I made sure that BellaBilling™ follows that philosophy closely.

The LLM performs reasoning and information extraction from natural conversation, while the surrounding workflow harness provides tools, orchestration, guardrails, and human oversight.

The intelligence comes from both the LLM and the architecture harness surrounding it.

System architecture

The workflow consists of eleven stages:

1. Customer calls Bella
2. BellaBilling™ answers
3. Conversation transcript
4. Invoice extraction
5. Google Sheets draft
6. Owner review
7. Approval gate within the client dashboard
8. Google Docs template
9. Custom-made invoice PDF generation
10. Email delivery
11. Customer receives invoice.

The entire workflow typically completes in under three minutes!

The most important stage is not the AI extraction. It is the approval gate.

Nothing is generated nor delivered until the business owner explicitly approves the invoice.

That design decision reflects one of the strongest themes throughout the course:

Architecture serves trust. Security serves the business.

Rather than maximizing automation, BellaBilling™ intentionally maximizes confidence.

The owner always remains responsible for customer-facing communication.

Applying Google AI intensive concepts

Building BellaBilling™ gave me an opportunity to apply several concepts introduced throughout the course:

Context engineering

Instead of relying on clever prompting alone, the system uses structured context gathered from a natural conversation to generate invoice drafts. The workflow focuses on providing the model with the information it needs rather than asking it to infer missing details.

Agent = Model + Harness

The model performs language understanding, but the surrounding workflow manages orchestration, state management, approvals, document generation, and delivery. The model is only one component of the complete system

Tool orchestration

BellaBilling™ coordinates multiple services throughout a single workflow.

Voice conversation is handled through Vapi.ai.

Workflow orchestration is managed through Make.

Google Sheets provides the owner review dashboard.

Google Docs generates the invoice.

Email delivery completes the workflow.

Each tool performs a specialized role within the overall architecture.

Human-in-the-loop design

One of the most important principles I wanted to preserve was human judgment.

AI **prepares** the work.

The owner **approves** the work.

The system **executes** the work.

Removing repetitive tasks should not remove accountability.

Security and Guardrails

Instead of allowing autonomous customer communication, BellaBilling™ introduces an explicit approval checkpoint before any invoice is generated or delivered.

This protects against accidental customer-facing mistakes while maintaining a fast workflow.

Deployability

Although developed as part of this capstone, BellaBilling™ was designed as a production-oriented workflow using real services rather than simulated components.

The demonstration video shows the complete workflow operating in a single, unedited take.

Building the system

The project began with a simple question:

"What would a billing coordinator actually do?"

Rather than starting with APIs or automation tools, I started by thinking through the real workflow a business owner follows after completing job.

That process naturally became the system architecture.

Only after the workflow was clear did I select the technologies needed to support each stage.

Throughout development, I found myself applying many of the principles from the course. Not because I was trying to satisfy competition requirements, but because they consistently led to better design decisions.

The biggest lesson was that building agents is less about generating text and more about designing trustworthy systems.

Lessons learned

Before taking Google's AI Intensive, I often thought about AI primarily in terms of prompting.

This course shifted my perspective towards architecture.

I learned that production systems depend on context engineering, orchestration, evaluation, guardrails, and thoughtful tool integration. It is not simply about choosing the best model.

That change influenced nearly every decision in BellaBilling™.

Instead of asking: **"Should AI do this?"**

The answer was often that AI should assist the business owner, not replace them.

That philosophy became the foundation of the entire workflow.

Conclusion

BellaBilling™ represents more than an invoicing tool…

It represents my attempt to apply the ideas from Google's AI Intensive to a practical business system that solves a real administrative problem.

By combining natural conversation, structured reasoning, workflow orchestration, and explicit human approval, BellaBilling™ demonstrates how AI agents can increase productivity while preserving trust and accountability.

This capstone has reinforced an idea that I expect to carry into every future product I build:

Architecture serves trust.

And when businesses trust the system, they can focus on serving their customers instead of managing paperwork.

Thank you for your time.